PROJECT DOCUMENTATION

Student Performance Analytics System — Python with AI Mini Project

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Technologies:	Python | NumPy | Pandas | Google Colab / Jupyter Notebook

PROJECT ABSTRACT

A comprehensive data-analysis project designed for evaluating student academic performance through dataset validation, subject-level analysis, department-level metrics, and automated reporting.

1. Project Title and Overview

Project Title: Student Performance Analytics System

The Student Performance Analytics System is a Python-based mini project developed to organize and analyze student academic data. It uses Pandas to load and process a CSV dataset and NumPy for numerical calculations. The program calculates student totals and averages, assigns grades, evaluates pass/fail status, identifies high and low performers, and summarizes performance by subject and department.

2. Objective

The main objective is to turn student marks and attendance records into clear, useful information. Reviewing each student's performance manually can take time and makes comparisons difficult. This project automates common calculations and presents results in a consistent format.

- Calculate total and average marks for each student.
- Assign grades using the grading rules implemented in the program.
- Determine pass/fail status using marks and attendance conditions.
- Identify students with the highest and lowest performance.
- Summarize subject-wise and department-wise performance.
- Review grade distribution and attendance.
- Export the processed results to a CSV report.

Problem Addressed

The project reduces repetitive manual calculations and makes it easier to inspect performance patterns across students, subjects, and departments. It is intended as an educational data-analysis tool, not as an automated decision system for high-stakes academic decisions.

3. Technologies Used

- **Python:** Implements the program logic, conditions, loops, reusable functions, file handling, and output messages.
- **Pandas:** Reads the CSV file into a DataFrame, displays tabular data, creates or processes analysis columns, groups records, and prepares the final report.
- **NumPy:** Supports numerical operations and summaries on marks and other numeric values.
- **Google Colab / Jupyter Notebook:** Provides an interactive environment for running the project in sections and reviewing output.
- **CSV:** Used as the input format for student records and as the output format for the processed report.

4. Dataset Description

Input File: student.csv

The dataset used for this project contains approximately 30 student records and 9 columns. The program reports the exact record and column counts when the file is loaded.

Column	Description
Student_ID	Unique identifier for each student
Name	Student name
Department	Academic department
Python	Python subject marks
SQL	SQL subject marks
HTML	HTML subject marks
CSS	CSS subject marks
JavaScript	JavaScript subject marks
Attendance	Student attendance percentage

Data note: The dataset should be checked for missing values, invalid marks, duplicate records, and values outside expected ranges before the results are interpreted.

5. Implementation

The notebook is arranged into numbered code sections, making it easier to run and review each stage of the analysis. The main workflow is described below.

Step 1 — Import libraries and load the dataset: The program imports os, NumPy, Pandas, and display utilities. It loads the student.csv file with Pandas read_csv() and displays the dataset, number of records, and number of columns.

Step 2 — Validate and prepare data: The dataset columns and values are checked before calculations are performed. Numeric subject and attendance fields are used for calculations. Validation helps detect data problems that could affect the summary.

Step 3 — Calculate totals and averages: For each student, the program calculates the total marks across Python, SQL, HTML, CSS, and JavaScript. Average marks are calculated from the same subject fields so students can be compared on a common scale.

Step 4 — Assign grades and pass/fail status: Grade assignment follows the thresholds defined in the notebook. The pass/fail condition used for this project requires at least 40 marks in every subject and at least 75% attendance. A student who does not meet either condition is classified as Fail.

Step 5 — Analyze performance: Pandas operations and reusable functions support summaries such as top and lowest performers, subject averages, department comparisons, grade distribution, and attendance analysis. Loops and conditional statements are used where repeated processing and decision-making are required.

Step 6 — Use NumPy and export results: NumPy is used for numerical calculations and summaries. The analyzed student records are exported to student_performance_analysis.csv for later review or submission.

6. Key Features

Student-Level Analysis

- Calculates each student's total and average marks.
- Assigns a grade based on the implemented grading criteria.
- Determines pass/fail status using subject marks and attendance.
- Identifies the highest- and lowest-performing students.

Group and Class Analysis

- Calculates subject-wise performance summaries.
- Compares results across departments.
- Summarizes grade distribution and attendance.
- Uses NumPy for numerical calculations.

Reporting

- Displays key analysis outputs in the notebook.
- Exports processed data to student_performance_analysis.csv.

7. Output / Screenshots

The notebook's output includes the loaded dataset, student totals and averages, grades and results, top/lowest performers, subject-wise summaries, department-wise summaries, grade distribution, attendance analysis, numerical summaries, and a completion message.

(1) Dataset Loaded Successfully With Record/Column Counts

```
STUDENT PERFORMANCE ANALYTICS SYSTEM
------------------------------------------------------------
Dataset loaded successfully!
Number of records: 30
Number of columns: 9
```

	Student_ID	Name	Department	Python	SQL	HTML	CSS	JavaScript	Attendance
0	101	Anjali	CSE	85	78	90	88	82	92
1	102	Rahul	CSE	72	80	75	70	78	85
2	103	Sneha	IT	95	92	88	94	96	98
3	104	Arjun	CSE	60	65	58	62	70	75
4	105	Priya	IT	88	85	91	89	90	94
5	106	Kiran	ECE	76	74	80	78	82	88
6	107	Meena	CSE	92	89	95	91	94	96
7	108	Vikram	IT	68	72	65	70	74	82
8	109	Divya	ECE	81	79	85	83	87	90
9	110	Rohit	CSE	55	62	58	60	65	78
10	111	Kavya	IT	90	94	89	92	91	97
11	112	Aditya	CSE	73	69	76	72	78	84
12	113	Neha	ECE	86	88	84	87	85	93
13	114	Manoj	IT	48	55	52	50	58	76
14	115	Pooja	CSE	79	82	80	77	84	89
15	116	Akash	ECE	64	60	68	65	70	80
16	117	Isha	IT	93	90	92	95	94	98
17	118	Varun	CSE	58	61	55	59	63	74
18	119	Riya	ECE	84	86	82	88	89	91
19	120	Sanjay	IT	70	75	72	74	76	86
20	121	Aarav	CSE	82	85	79	84	88	91
21	122	Simran	IT	91	87	93	90	89	95
22	123	Harish	ECE	67	71	69	73	75	83

	Student_ID	Name	Department	Python	SQL	HTML	CSS	JavaScript	Attendance
23	124	Nisha	CSE	89	92	86	88	91	96
24	125	Varun	IT	74	78	72	76	80	87
25	126	Swathi	ECE	96	94	91	93	95	99
26	127	Manish	CSE	63	68	61	65	70	79
27	128	Keerthi	IT	85	83	88	86	90	92
28	129	Rakesh	ECE	77	75	81	79	84	89
29	130	Aditi	CSE	94	91	96	92	93	97

(2) Student Performance Table With Totals, Averages, Grades And Results

```
======================================================================
STUDENT PERFORMANCE RESULTS
======================================================================
```

	Student_ID	Name	Department	Total_Marks	Average_Marks	Grade	Result	Attendance	Performance
0	101	Anjali	CSE	423	84.6	A	Pass	92	Very Good
1	102	Rahul	CSE	375	75.0	B	Pass	85	Very Good
2	103	Sneha	IT	465	93.0	A+	Pass	98	Excellent
3	104	Arjun	CSE	315	63.0	C	Pass	75	Good
4	105	Priya	IT	443	88.6	A	Pass	94	Very Good
5	106	Kiran	ECE	390	78.0	B	Pass	88	Very Good
6	107	Meena	CSE	461	92.2	A+	Pass	96	Excellent
7	108	Vikram	IT	349	69.8	C	Pass	82	Good
8	109	Divya	ECE	415	83.0	A	Pass	90	Very Good
9	110	Rohit	CSE	300	60.0	C	Pass	78	Good
10	111	Kavya	IT	456	91.2	A+	Pass	97	Excellent
11	112	Aditya	CSE	368	73.6	B	Pass	84	Good
12	113	Neha	ECE	430	86.0	A	Pass	93	Very Good
13	114	Manoj	IT	263	52.6	D	Pass	76	Average
14	115	Pooja	CSE	402	80.4	A	Pass	89	Very Good
15	116	Akash	ECE	327	65.4	C	Pass	80	Good
16	117	Isha	IT	464	92.8	A+	Pass	98	Excellent
17	118	Varun	CSE	296	59.2	D	Fail	74	Average
18	119	Riya	ECE	429	85.8	A	Pass	91	Very Good
19	120	Sanjay	IT	367	73.4	B	Pass	86	Good
20	121	Aarav	CSE	418	83.6	A	Pass	91	Very Good
21	122	Simran	IT	450	90.0	A+	Pass	95	Excellent
22	123	Harish	ECE	355	71.0	B	Pass	83	Good
23	124	Nisha	CSE	446	89.2	A	Pass	96	Very Good
24	125	Varun	IT	380	76.0	B	Pass	87	Very Good
25	126	Swathi	ECE	469	93.8	A+	Pass	99	Excellent
26	127	Manish	CSE	327	65.4	C	Pass	79	Good

	Student_ID	Name	Department	Total_Marks	Average_Marks	Grade	Result	Attendance	Performance
27	128	Keerthi	IT	432	86.4	A	Pass	92	Very Good
28	129	Rakesh	ECE	396	79.2	B	Pass	89	Very Good
29	130	Aditi	CSE	466	93.2	A+	Pass	97	Excellent

(3) Subject/Department Analysis

```
======================================================================
SUBJECT-WISE PERFORMANCE
======================================================================
```

	Subject	Average_Marks	Highest_Marks	Lowest_Marks	Students_Passed	Students_Failed
0	JavaScript	82.03	96	58	30	0
1	CSS	79.00	95	50	30	0
2	SQL	78.67	94	55	30	0
3	HTML	78.37	96	52	30	0
4	Python	77.83	96	48	30	0

```
Best subject: JavaScript (82.03)
Lowest-average subject: Python (77.83)
```

(4) Final CSV Export or Project Completion Output.

Analysis exported successfully: student_performance_analysis.csv

or

STUDENT PERFORMANCE ANALYTICS SYSTEM COMPLETED

1. Dataset loaded and validated.
2. Student totals, averages, grades, and results calculated.
3. Subject-wise analysis completed.
4. Top-performing students identified.
5. Department-wise analysis completed.
6. Grade distribution and attendance analyzed.
7. NumPy calculations completed.
8. Final CSV file generated.

Thank you!

8. Final Outcome

The project provides a repeatable workflow for converting student records into performance summaries. It brings together core Python programming, Pandas data handling, and NumPy numerical operations in one notebook.

When the notebook runs successfully with the dataset, it produces calculated student metrics, pass/fail classifications, subject and department summaries, attendance and grade analysis, and an exported CSV report. Exact counts and numerical findings should be taken from the notebook's actual output after the final run.

9. Challenges and Learning

Challenges

- **Dataset filename and file location:** The notebook must load the intended file, student.csv, from the working environment.
- **Data quality:** Incorrect or missing marks and attendance values can affect calculations and should be checked.
- **Consistent calculations:** Totals, averages, grades, and pass/fail status must use the intended subject columns and rules.
- **Interpreting results:** Summary statistics describe the dataset but do not automatically explain why a student performed a certain way.

How These Were Addressed

The loading section uses the specific dataset filename and prompts for an upload in Google Colab if it is not found. The analysis is organized into numbered sections, and explicit grading and pass/fail rules make the logic easier to review. The output CSV allows results to be checked after execution.

Learning Outcomes

- Learned how to load CSV data into a Pandas DataFrame.
- Practiced variables, data types, loops, functions, and conditions.
- Used NumPy and Pandas for numerical and tabular analysis.
- Learned to summarize results and export processed data.
- Understood the importance of validating data and explaining analysis limitations.

10. GitHub Repository and Submission

GitHub Repository URL:

> **REPOSITORY LINK**
> https://github.com/Yasasvi2025/Mini-Project-Student-Performance-Analytics-System-Python-with-AI

The repository should contain the project notebook or Python source file, the input dataset (where sharing is permitted), README.md, and this project documentation. After uploading, open the repository page and confirm that the files are visible and that the notebook runs with the submitted dataset.

Final Submission Checklist

- Project notebook/source code uploaded.
- student.csv dataset included, where permitted.
- README.md added.
- Project documentation added.
- Notebook tested from start to finish.
- Generated CSV output checked.
- GitHub URL verified and added above.

Conclusion

The Student Performance Analytics System demonstrates how Python, NumPy, and Pandas can be applied to a practical academic dataset. It automates common calculations, supports comparisons across subjects and departments, evaluates results using stated rules, and creates a reusable CSV report. The project also provides hands-on practice in programming fundamentals, data validation, analysis, and documentation.